SOIL EROSION AND CONSERVATION · NOTE CM04

Bunds and Bench Terraces — Graded Bunds, Bench Terraces, Contour Bunds and Waste Weirs

Course topics: SWCE4.5 L10, SWCE4.5 L11, SWCE7.4 L8, SWCE-510 T8

Graded bunds

- Common in India and comparable to narrow-base terraces. They dispose of excess runoff safely in high-rainfall areas with impervious soils; bunds and their channels are not farmed.
- **Uniform graded bund:** short bunds with small discharge, uniform grade throughout. **Variable graded bund:** longer bunds with larger discharge, grade varied by sections to keep velocity non-erosive.
- **Design** fixes the dimensions and the slope of the disposal channel for a non-erosive velocity, by either (1) a parabolic channel plus an earthen bund (for steeper slopes), or (2) a bund on a suitable grade with water flowing behind it (for gentler slopes — better suited to Indian farming).
- **Criteria:** vertical interval as for graded terraces; channel capacity by the rational method; dimensions by Manning's formula; freeboard 15 cm or 20% of design depth, whichever is more.
- **Section:** set by side slopes, height, base width and top width. Height not less than 45 cm; top width 30–90 cm (wider for taller bunds); base wide enough that the seepage line does not emerge above the downstream toe.
- **Construction:** by manual labour, animal-drawn buck scraper, bulldozer or motor grader, with soil from the channel or, if there is none, scraped from both sides.

Soil	Minimum stable side slope (H : V)	Soil	Seepage-line slope (H : V)
Clayey	1 : 1	Clayey	3 : 1
Loamy	1.5 : 1	Sandy loam	5 : 1
Sandy	2 : 1	Sandy	6 : 1

Example 14.3 — data given for a 350 m graded bund
Sandy loam; land slope 3%; horizontal spacing 70 m; cultivated crop
Channel grade 0.1% (first 100 m), 0.12% (next 100 m), 0.14% (rest)
1-hour rainfall for a 10-year recurrence: 10 cm/hr
Peak runoff by the rational method: Q = 0.0276·C·I·A (I in cm/hr, A in ha)
Time of concentration: Tc = 0.0195·L^0.77·S^(-0.385)
The notes stop at the data; the method is: catchment per bund = 350 × 70 m = 2.45 ha → Tc → intensity for Tc → Q → section by Manning's formula for each grade reach.

Corrections: the notes write the Kirpich exponent as S^0.385; it is S^(-0.385) (Tc falls as slope rises). The rational constant with I in cm/hr is 1/36 = 0.0278; 0.0276 is a rounded version.

Bench terraces

- One of the oldest erosion-control methods for cultivating steep land, widely used in the hills of Kerala, Karnataka, Himachal Pradesh, Uttarakhand, the North-East and the Nilgiris (Tamil Nadu).

- Steep land is turned into a series of level or nearly level steps across the slope, separated by almost vertical risers — supported by masonry (usually stone), or of earth if sloping. Generally used on slopes steeper than 15%, where cultivable land is scarce and the cost and farming problems are justified.

Type	Suits
Level (table-top)	Medium rainfall, permeable deep soils
Sloping outward	Low rainfall, permeable soils of medium depth
Sloping inward (most common)	Heavy rainfall — more effective

Design

- Excavated bench terraces use the **cut-and-fill principle**: the vertical interval is chosen so the area cut equals the area filled. Make risers as steep as possible — vertical with masonry, at the angle of repose with earth (steeper with vegetative stabilisation).
- The design covers spacing, grade and length, and cross-section. Spacing, as the vertical interval, sets the depth of cut, the height of fill and the total drop; soil and slope limit it. Less cutting keeps more productive soil depth, and the bench must be wide enough for convenient farming.

With land slope S (%), vertical drop D and bench width W, and a 1 : 1 riser:

$$\text{Horizontal distance for drop } D = 100D/S = W + D$$

$$\rightarrow D = W \cdot S/(100 - S)$$

Illustration (not in the notes)
S = 20%, W = 4.5 m → D = 4.5 × 20/(100 - 20) = 1.125 m

- **Nilgiris:** minimum width 4.5 m on 15–25% slopes and 3 m on 25–33% slopes (33% is the maximum slope recommended for agricultural crops).
- **Gradient:** benches must shed excess runoff at non-erosive velocity; longitudinal slopes up to 0.75% are safe. In the common inward-sloping type the inner edge is ≈ 30 cm lower than the outer edge, and a small channel carries excess rain to a grassed or forested area or a protected ditch.
- **Runoff:** estimate the critical runoff by the rational method and set the channel slope by Manning's formula so velocity stays non-erosive. Without research data, the maximum bench length in the Nilgiris is 105 m.
- **Cross-section:** earth cut from the upper half is deposited on the lower slope as the fill. As slope increases stability falls, so key the fill to the original ground — key trenches, vegetation removed and the surface loosened (preferably disced) before filling.
- **Shoulder bund** (outer edge), stable against slipping and overtopping: freeboard ≥ 15 cm, side slopes 1 : 1, height ~30 cm, bottom width 75 cm.
- **Alignment:** consider tillage convenience and field boundaries; keep the minimum convenient width (adjusting near boundaries), smooth sharp curves, and depart from the contour where necessary.
- **Construction:** finalise alignment, prepare the embankment site (plough, clear vegetation, dig key trenches if needed), push the excavated earth downslope until the level is reached — with animal-drawn buck scraper and float, hand tools or bulldozer. Shape shoulder bunds by hand and protect edges with deep-rooted grasses. Stone-walled bench terraces with near-vertical sides are an age-old practice, justified where stone is cheap — usually for perennial tree crops, but usable for annual crops.

Example 14.4 (partial data in the notes)
150 m bench terrace, average slope 20%
Manning: V = R^(2/3)·S^(1/2)/n ≈ 0.497 m/s (≈ 50 cm/s)
Q = a × v = 0.609 × 0.497 = 0.3028 m³/s — the section has enough capacity and the velocity is non-erosive.
(Check: 0.609 × 0.497 = 0.3027.)

Contour bunds

- An earthen embankment on the contour. Small bunds across the slope cut a long slope into strips; each bund is a barrier that reduces the amount and velocity of runoff — comparable to narrow-base level terraces. Widely practised in Maharashtra, Gujarat, Tamil Nadu, Karnataka and Andhra Pradesh.
- **Functions:** divide the area into strips, block runoff, reduce velocity and erosion, improve infiltration and raise groundwater storage.
- **Peripheral bunds** run along field boundaries regardless of the contour. They serve as fences and protect against water and wind erosion in low-rainfall areas, but in heavy rain on low-infiltration soils they give way and accelerate erosion.
- No cultivation on the embankments — about 5% of the land is lost under bunds.

Indian experience

- Contour bunding in India rests largely on empirical rules, experience and broad engineering principles. The former Bombay State (Maharashtra and Gujarat) pioneered it, bunding more than 5 lakh hectares.
- Bunds succeeded on shallow, medium and medium-deep soils but **failed on deep black soils** — cracking in hot weather, caking in the monsoon, water percolating through cracks, breaches, poor drainage, collapsing sides and grasses that do not establish. They also fail on very shallow soils (< 7.5 cm).

Soil class (Gadkary, 1954)	Definition
Shallow	Depth < 22.5 cm
Medium	22.5–45 cm
Medium deep	45–90 cm
Alluvial deep	Clay ≤ 30%
Deep heavy	Depth > 90 cm and/or clay > 30%

Slope (%)	Vertical interval (m)	Approx. horizontal distance (m)
0–1	1.05	105
1–1.5	1.20	98
1.5–2	1.35	75
2–3	1.50	60
3–4	1.645	52

Table 14.10 — spacing of contour bunds.

Soil (Table 14.11)	Base width (m)	Top width (m)	Height (m)	Side slope	Area (m²)
Shallow (7.5–22.5 cm)	2.67	0.38	0.75	1½ : 1	1.14
Medium (22.5–45 cm)	3.12	0.60	0.85	1½ : 1	1.56
Medium deep (45–90 cm)	4.25	0.60	0.90	2 : 1	2.18

Check: the trapezoid areas (base + top)/2 × height are 1.14, 1.58 and 2.18 m², so the medium-soil area printed as 1.56 m² is slightly low.

Design

- Choose the vertical and/or horizontal interval and the cross-section, on the same principles as terraces and graded bunds; without detailed data use the recommended tables.
- **Height:** enough storage above the bund for the maximum runoff from a 10-year recurrence rainfall; freeboard about 20% of design depth after settlement.

$$\text{Cross-section area } (\text{m}^2) = \text{runoff (cm)} \times \text{bund horizontal interval (m)} \,/\, 100$$

- **Allowable submergence:** the waste-weir level and the submerged area depend on the crop, the soil's infiltration rate and the duration of ponding.
- **Moisture conservation:** on paddy land store all rain (no runoff); for other crops base capacity on average consumptive use (cm/day) and the longest dry spell, with the waste weir set to release the excess.
- **Economy:** minimise earthwork cost plus the value of land lost permanently.
- **Critical length:** velocity grows with steepness, slope length and contributing area; space bunds so velocity stays non-erosive — very close spacing raises cost and narrows fields.
- **Seepage:** allow for seepage through the bund, which depends on head, side slopes and soil permeability.
- **Length and alignment** follow broad-base terraces; lay the upstream toe line with surveying instruments.

Construction

- By manual labour, animal power (the buck scraper is most efficient), bulldozers or motor graders.
- **Borrow pits** on the upslope side, wide and shallow, with a 3 m berm between the bund base and the pit; they fill in 2–3 years and the land between bunds is cultivated.
- **With a buck scraper:** plough or disc to loosen the soil and scarify for bonding; mark the base lines with a mouldboard plough, throwing the furrow slice toward the bund (resetting stakes if furrows fill); scrape soil from both sides and dump it on the strip between the base lines — a lead of about 6 m on either side works well; shape by hand and recheck the top levels with a level. This avoids nearby borrow pits and cuts cost.

Outlets and waste weirs

Bunds seldom hold all the rain, so outlets are usually needed — except on permeable soils in low-rainfall areas.

Waste weir	Description
Clear overfall weir	Like a drop spillway, usually brick or stone masonry in cement mortar and designed the same way; the crest of the weir notch is about 30 cm above ground
Channel weir	Grassed channel at the end of the bund, with the sill about 30 cm above ground and an underground stone wall
Pipe outlet	Like a drop-inlet spillway, using reinforced-concrete, vitrified sewer or corrugated-metal pipe: 15 cm pipe for a 3-hectare catchment, 22.5 cm for about 5 hectares, 30 cm for up to 8 hectares. Inlet crest 30 cm above ground

Sources: ICAR eCourse 'Soil and Water Conservation Engineering' — Dr A. Mishra, reviewed by Dr B. C. Mal (IIT Kharagpur); AgriMoon e-book. Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. Minimum height of a graded bund is

(a) 30 cm (b) 45 cm
(c) 60 cm (d) 90 cm

Q2. Freeboard for graded bunds is

(a) 10 cm (b) 15 cm or 20% of depth, whichever is more
(c) 25% of depth (d) 30 cm

Q3. Seepage-line slope (H : V) in sandy soil is

(a) 3 : 1 (b) 4 : 1
(c) 5 : 1 (d) 6 : 1

Q4. Bench terraces are generally adopted on slopes steeper than

(a) 3% (b) 8%
(c) 15% (d) 33%

Q5. For a 1 : 1 riser, the vertical drop between bench terraces is

(a) WS/100 (b) WS/(100 - S)
(c) W(100 - S)/S (d) 100W/S

Q6. Bench terraces for heavy rainfall should slope

(a) outward (b) inward
(c) level (d) downstream

Q7. Contour bunds failed mainly in

(a) shallow soils (b) medium soils
(c) deep black soils (d) alluvial soils

Q8. Area lost under contour bunds is about

(a) 1% (b) 5%
(c) 15% (d) 25%

Q9. A 22.5 cm pipe outlet serves a catchment of about

(a) 1 ha (b) 3 ha
(c) 5 ha (d) 8 ha

Q10. Contour bund cross-section (m^2) =

(a) runoff (cm) × H.I. (m)/100 (b) runoff × V.I.
(c) H.I./runoff (d) $runoff^2/100$

Answers

Q1 — (b) 45 cm

Q2 — (b) 15 cm or 20% of depth, whichever is more

Q3 — (d) 6 : 1

Q4 — (c) 15%

Q5 — (b) WS/(100 - S)

Q6 — (b) inward

Q7 — (c) deep black soils

Q8 — (b) 5%

Q9 — (c) 5 ha

Q10 — (a) runoff (cm) × H.I. (m)/100